CRAB: A NOVEL CLUSTERING SCORE USING CLUSTERING WITH RIVALS AND BUDDIES FOR UNSUPERVISED LEARNING

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by

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ABSTRACT

cRab: A Novel Clustering Score using Clustering with Rivals and Buddies for Unsupervised Learning

Allen Choi

We present a novel clustering score approach relying on a method that classifies pairs of points as either ”rivals” or ”buddies”, constructing a score for a given number of clusters. We then present results demonstrating the properties of this metric on datasets with no clusters.

Keywords: Clustering. Unsupervised Clustering. Score. Machine Learning.

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TABLE OF CONTENTS

Page

LIST OF TABLES vii

LIST OF FIGURES viii

CHAPTER

1 The Problem and What's been Tried 1

1.1 Motivation 1

1.1.1 Elbow Method 1

1.1.2 Silhouette Score 3

1.2 Literature Review 4

2 Details 5

2.1 Calculated 5

2.2 Performance on Simulated Data 5

2.3 Sensitivity 5

2.4 Known Data 5

2.5 Performance on penguins dataset 5

3 Computational Details 7

3.1 Matrix Stacking 7

3.2 Algorithm Construction 8

3.3 Score Construction 8

3.4 Distribution of Ones and Neg Ones 9

3.5 Speed and Memory Scaling 9

4 Null Distribution 11

4.1 Null Distribution of Scores 11

4.2 Null Distribution over Resamples 12

5 Conclusion 13

REFERENCES 14

APPENDICES

A Resample Matrix 15

B Mean Matrix 19

C Metric Calculation 20

D Point Contribution Function 21

E Get Point Vector Helper 22

LIST OF TABLES

Table Page

4.1 Variances for 20% Resample Calculation 12

LIST OF FIGURES

Figure		Page
1.1	Scatterplot of the Example Dataset	2
1.2	WCSS as Number of Clusters k Increases	3
1.3	Silhouette Score as Number of Clusters k Increases	4
2.1	Boxplots of Simulated Scores Across k On Penguins Dataset	5
3.1	Memory and Speed of the cRab Algorithm as Number of Observations Increases	9
3.2	Memory and Speed of the cRab Algorithm as Number of Observations Increases	10
4.1	Distribution of Scores on a Random p-Dimensional Normal Distribution, Faceted by Resample %	11

Chapter 1

THE PROBLEM AND WHAT'S BEEN TRIED

1.1 Motivation

Unsupervised clustering algorithms such as K-Means and hierarchical clustering are commonly used today when the number of clusters in a given dataset is not known beforehand, differentiating it from supervised clustering. Kodinariya (2013) presents the most common approaches used, such as the elbow method and Silhouette score. However, even with these scores there are often disagreements and caveats.

1.1.1 Elbow Method

The elbow method (Thorndike, 1953) is a relatively simple technique that utilizes the plotting of the sum of the Within Cluster Sum of Squares (WCSS) across all clusters at the y axis, against the number of clusters on the x-axis. Similar in behavior to the R^2 measure, adding more clusters necessarily decreases the WCSS due to the fact that adding an extra cluster will mean that at least one observation is closer to that cluster than the others. The question of "how many clusters" is then answered by looking at where this drop-off in WCSS is significant. This method however comes with many problems (Schubert, 2023). The biggest issue is that while for some datasets this "drop-off" in WCSS is easy to visually recognize, sometimes it can be extremely difficult to tell what is considered significant. In fact, (Schubert, 2023) takes a strong approach on this, where they state "Educators should omit the method or at least explain better alternatives".

We demonstrate this below with an example dataset, where

$$x_1, y_1 \sim exp(\lambda = 1)$$

$$x_2, y_2 \sim exp(\lambda = 3) + 1$$

$$x_3, y_3 \sim exp(\lambda = 5) + 2$$

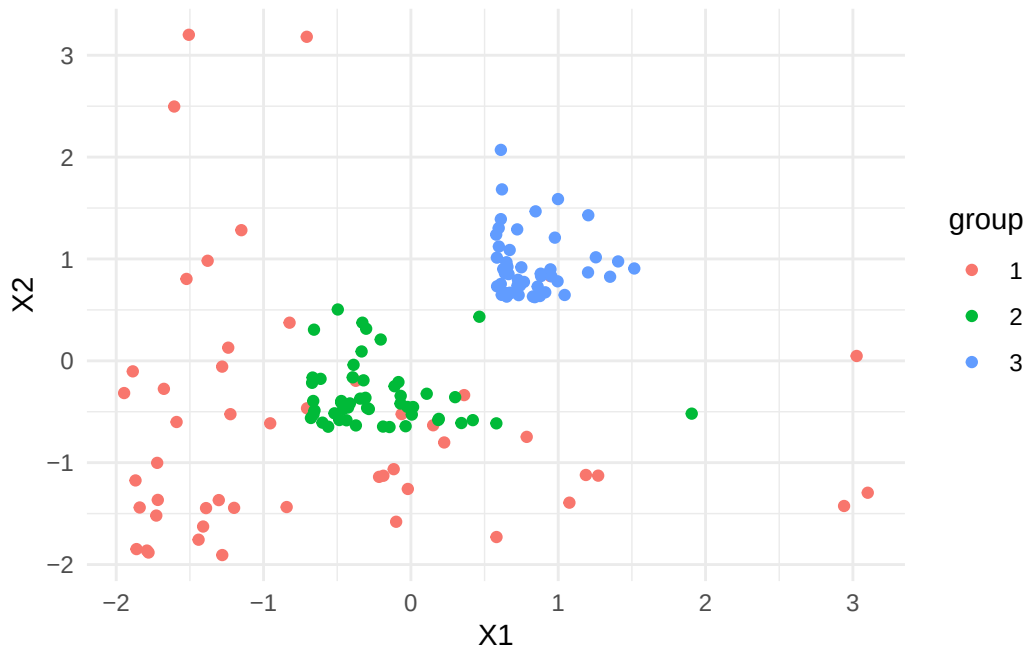


Figure 1.1: Scatterplot of the Example Dataset

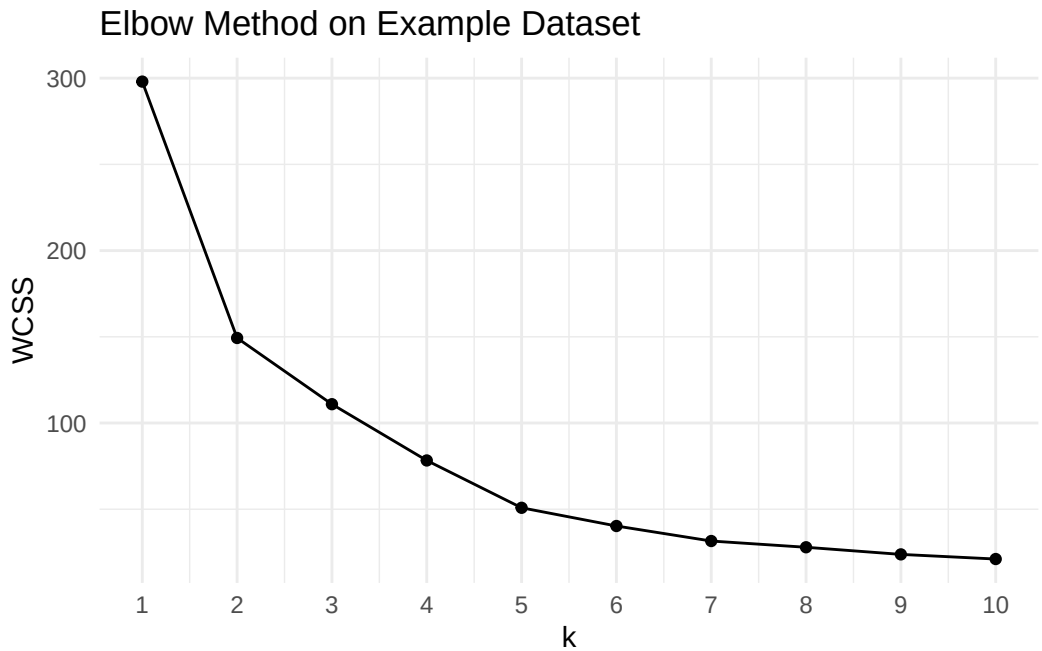


Figure 1.2: WCSS as Number of Clusters k Increases

1.1.2 Silhouette Score

The Silhouette Score approach (Rousseeuw, 1987) is a method proposed to measure how comfortably each observation belongs to its assigned cluster compared to the other possible clusters. As it is defined in the original 1986 paper, the silhouette width $s(i)$ for observation i is defined by:

$$s(i) = \frac{b(i) - a(i)}{max\{a(i), b(i)\}}$$

where $a(i)$ is defined by the average distance between the i and all other observations in the same cluster, and $b(i)$ is defined by the smallest average distance between i and all other observations in each other respective clusters.

Averaging this score gives a measure of how well a certain number of k-Clusters fits the data, similar to the elbow method. Unlike in the elbow method however, where increasing the number of clusters necessarily decreases the WCSS, the average Silhouette width does

not necessarily increase as the number of clusters increases.

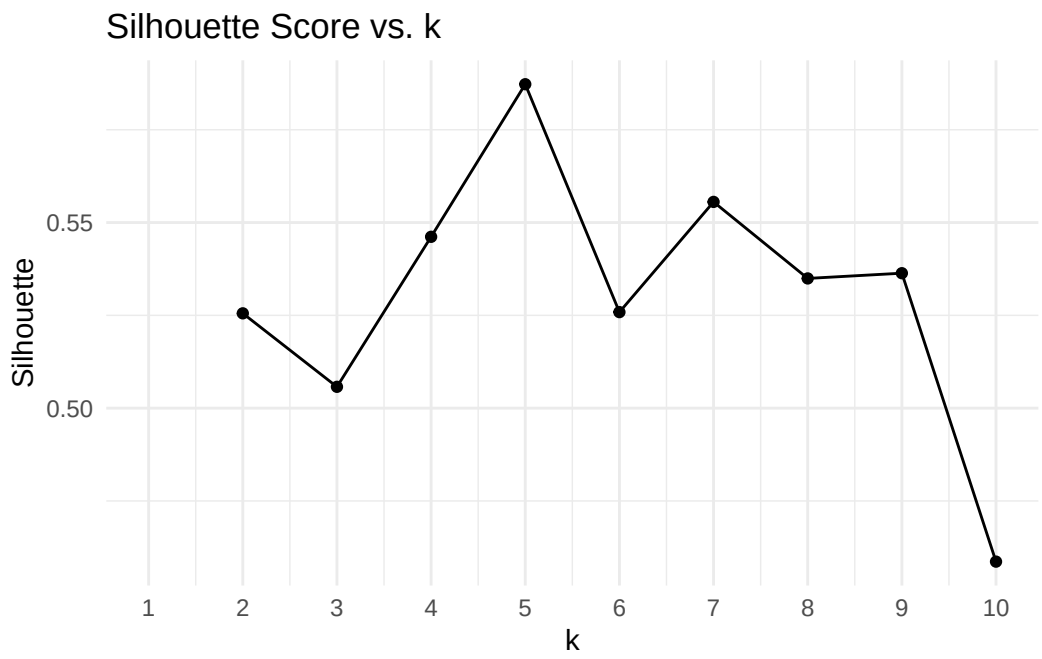


Figure 1.3: Silhouette Score as Number of Clusters k Increases

Of course, it seems disingenuous to not include an example of where these methods work perfectly well:

We present the cRab algorithm, which utilizes random subsampling of a dataset in order to determine the optimal number of clusters for an unsupervised clustering algorithm.

(R Core Team, 2025). (Ben-Hur, Elisseeff, & Guyon, 2001). (Jaeger & Banks, 2023). (Kuhn & Wickham, 2020). (Hvitfeldt & Bodwin, 2025). (Wickham et al., 2019).

1.2 Literature Review

Chapter 2

DETAILS

2.1 Calculated

2.2 Performance on Simulated Data

2.3 Sensitivity

2.4 Known Data

2.5 Performance on penguins dataset

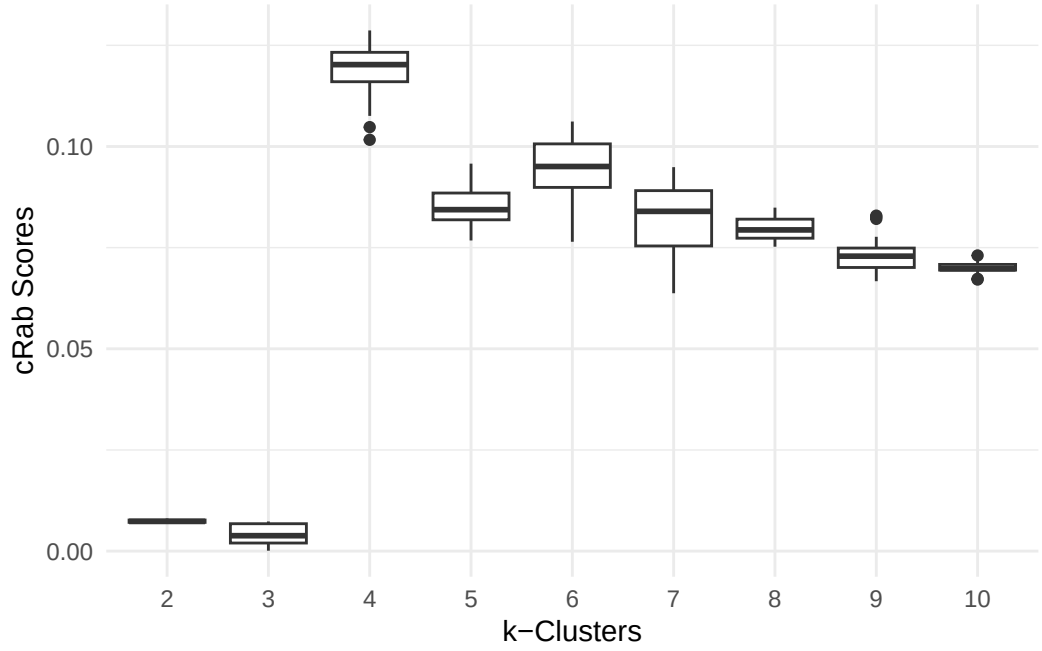


Figure 2.1: Boxplots of Simulated Scores Across k On Penguins Dataset

The cRab algorithm was ran 10 times on each of the k-Clusters, randomizing the starting seed

Chapter 3

COMPUTATIONAL DETAILS

The main principle of our algorithm works off of is the fact that subsamples of a dataset should still capture the overall clustering structure, which is also echoed in the paper from Ben-Hur (2002). This means that given two distinct points and their cluster assignments across resamples, the clustering assignments should be consistent. For example, two points that are far away from each other should consistently not be in the same cluster, and two points that are close together should consistently be in the same cluster, even across resamples.

3.1 Matrix Stacking

We now need a framework for the handling of resampling the matrices.

A "resample matrix" for each resample is then needed, where for points i and j we have a value of 1 in the matrix if they belong to the sample cluster, -1 if not, and 0 if either point was not in the resample.

So, for two points i, j in the resample matrix, the element p_{ij} of the resample matrix is defined as:

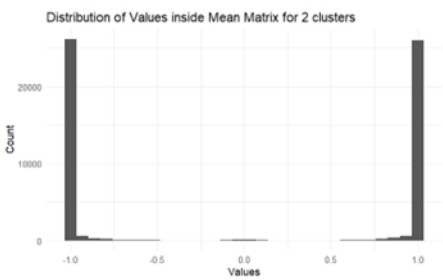
$$p_{ij} = \begin{cases} 1, & \text{if points i, j belong to the same cluster} \\ -1, & \text{if points i, j do not belong to the same cluster} \\ 0, & \text{if either point i or point j are not in the resample} \\ \text{NA}, & \text{if pairwise element ij is located in the lower triangle or diagonal} \end{cases}$$

3.2 Algorithm Construction

After we have n matrices, we then reduce the matrices, taking the mean with respect to each index pair, ignoring zeroes. We then receive a final "mean matrix" of size m x m, where m is the number of observations in the dataset.

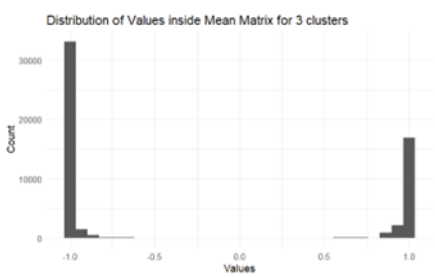
3.3 Score Construction

To illustrate the construction of the final score given this mean matrix, we use the penguins dataset in R with three known species and scaled bill length and flipper lengths.



The distribution of values inside the mean matrix after running the algorithm on the penguins dataset, using K-Means and k=2

(replace with code that generates this later)



(replace with code that generates this later)

(Also remove the images folder later)

Figure 2: The distribution of values inside the mean matrix after running the algorithm on the penguins dataset, using K-Means and k=3

We see that from Figures 1 and 2 that the distribution of values inside the mean matrix changes depending on the number of clusters set beforehand in the K-Means algorithm, albeit slightly. For k = 2, there are a small number of points around the value of 0, which indicates that these specific points are ones who are changing "allegiances" often, meaning that they switch from being "buddies" (being in the same cluster) and to being "rivals", (not being in the same cluster).

Then, to make a final score taking all this into account, we decided to vectorize this matrix, then taking the absolute value and getting the sum of squared distances from 1. This is in order to heavily penalize values that are farther from 1.

3.4 Distribution of Ones and Neg Ones

3.5 Speed and Memory Scaling

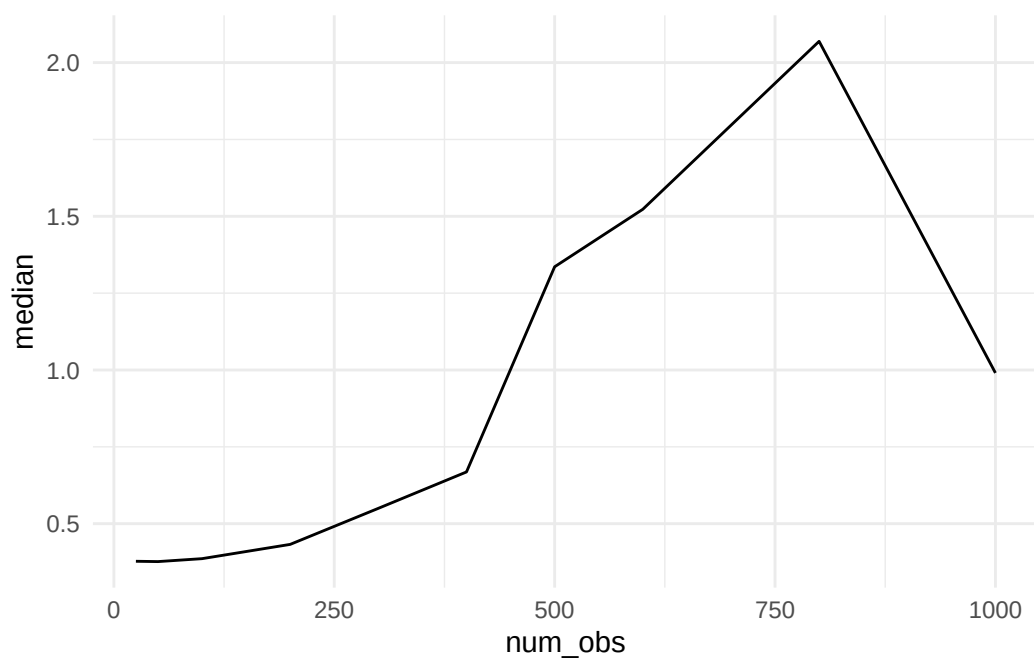


Figure 3.1: Memory and Speed of the cRab Algorithm as Number of Observations Increases

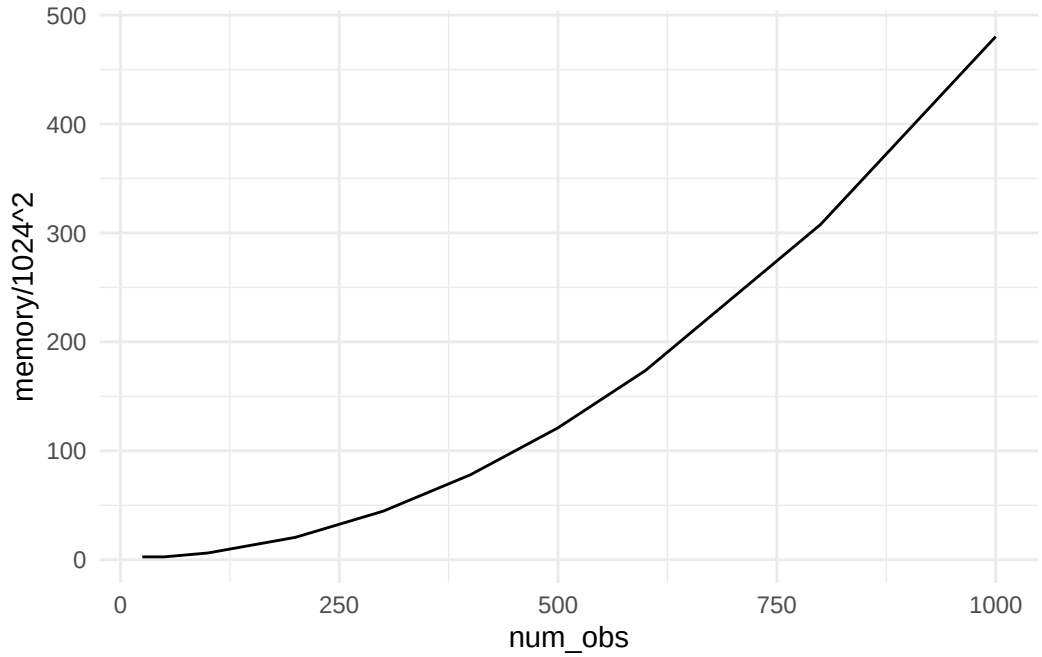


Figure 3.2: Memory and Speed of the cRab Algorithm as Number of Observations Increases

Constant up until a certain point, after which it becomes quadratic in terms of both memory and time.

Chapter 4

NULL DISTRIBUTION

4.1 Null Distribution of Scores

It is also interesting information to see what the distribution of scores looks like when there are no clusters in the dataset. This is so that our scores can be adjusted depending on this "null distribution", also allowing us to get a degree of evidence for a given k and other parameters.

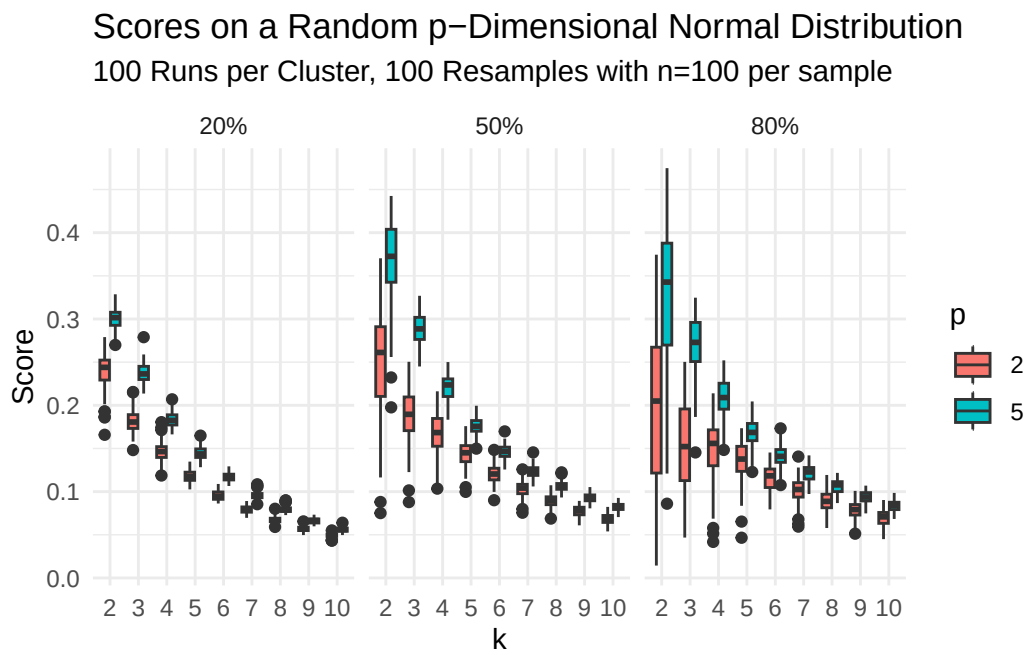


Figure 4.1: Distribution of Scores on a Random p-Dimensional Normal Distribution, Faceted by Resample %

We see in the above figure that the resample proportion has a significant impact on the distribution of scores, where the variability and mean of scores over many simulation runs is dependent on the resample proportion. The scores also tend to be higher, scaling with

p-dimensions, with a decreasing score distribution as k increases. Notably, all these scores look symmetrically distribution around their mean, with respect to each k.

4.2 Null Distribution over Resamples

```
`summarise()` has grouped output by 'factor(k)'. You can override using the
`.groups` argument.
```

Table 4.1: Variances for 20% Resample Calculation

factor(k)	factor(prop_resample)	var
2	0.2	0.1347830
3	0.2	0.1471547
4	0.2	0.1285054
5	0.2	0.1161971
6	0.2	0.1133949

Chapter 5

CONCLUSION

On the penguins dataset, we use 100 resamples and a resample proportion of 80% wih the k-Means algorithm and a given k, using again the scaled bill length and flipper lengths in our dataset.

(i guess this should also go in the analysis portion??)

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APPENDICES

Appendix A

RESAMPLE MATRIX

```
resample_function <- function(data = data,
                              formula = ~ .,
                              k = 3,
                              number_of_resamples = 15,
                              proportion_resample = 0.9,
                              starting_seed = 599,
                              algorithm = "kmeans") {
  data <- data |>
    drop_na()
  data$index <- 1:nrow(data)
  results <- list()

  # For loop
  results <- foreach(i = 1:number_of_resamples,
                     .packages = c("tidyclust",
                                   "dplyr",
                                   "tidymodels")) %dopar% {
    # Reproducibility over parallel
    set.seed(starting_seed + i)
    result_matrix <- matrix(0, nrow = nrow(data),
```

```
                          ncol = nrow(data))


random_sample <- data |>
  filter(index %in% sample(index,
                           proportion_resample * nrow(data)))


if (algorithm == "kmeans") {
  cluster_assigned <- k_means(num_clusters = k) |>
    fit({{formula}},
        data = random_sample |>
          select(-index))
} else if (algorithm == "hier_clust") {
  cluster_assigned <- hier_clust(num_clusters = k) |>
    fit({{formula}},
        data = random_sample |>
          select(-index))
} else {
  stop("Algorithm is not supported. Please select
       one of (kmeans, hier_clust)")
}



intermediate <- data.frame(random_sample$index,
                           extract_cluster_assignment(
                             cluster_assigned
                             ) |>
                             mutate(.cluster = as.character(
```

```
                                .cluster)
                                ),
                            stringsAsFactors = FALSE)
colnames(intermediate) <- c("index", "cluster")

for (c in unique(intermediate$cluster)) {
  idx <- intermediate[intermediate$cluster == c, ]$index

  # Check that the index list is not length 1 for
  # purposes of n > m for combn
  # This also fixes a bug in the
  # previous code where idx was wrongly
  # interpreted as single numerical
  # value as param inside combn
  if (length(idx) > 1) {
    idx <- sort(idx)
    ones <- t(combn(idx, 2))
    result_matrix[cbind(ones[, 1], ones[, 2])] <- 1
  }

  # Note for future self: This calculates the indices, even across y=x line
  # which means we are doing double the work for this part.
  neg_one_idx <- expand.grid(idx, setdiff(random_sample$index, idx))
  result_matrix[cbind(neg_one_idx[, 1], neg_one_idx[, 2])] <- -1
}
result_matrix[lower.tri(result_matrix, diag = TRUE)] <- NA
result_matrix
```

```
    }
  return(results)
}
```

Appendix B

MEAN MATRIX

```r
mean_matrix <- function(list_of_matrices = list_of_matrices) {
  # Absolute value the entire matrix, for all matrices
  absolute_sum <- lapply(list_of_matrices, abs)

  # Get the sum of absolute values in the final matrix
  absolute_final <- absolute_sum |>
    reduce(`+`)

  # Get the sum of values in the final matrix
  numerator <- list_of_matrices |>
    reduce(`+`)

  # Returns the matrix of zero-removed means.
  # The zero-removed mean is the sum over
  # the (length - the number of zeroes).
  # The length - the number of zeroes is
  # equivalent to the sum over absolute values
  return(numerator / absolute_final)
}
```

Appendix C

METRIC CALCULATION

```
squared_distance_from_one <- function(mean_matrix = mean_matrix) {
  res_vec <- as.vector(mean_matrix) # Ignore the order of the elements in matrix
  res_vec <- abs(res_vec[!is.na(res_vec)]) # Absolute value and remove NAs
  return(mean((1 - res_vec)^2)) # Take average squared distance from one
}
```

Appendix D

POINT CONTRIBUTION FUNCTION

```
point_contribution <- function(data, mean_matrix) {
  # Given the original data (the dataset inputted into the resample_matrices funct
  # and the mean_matrix, returns each observation's contribution to the cRab Score
  # We don't necessarily require the second arg to be the mean matrix, for example
  # it could be resample matrix j.
  # This is returned as an (index, score) dataframe.

  # Get a list of lists containing index and index "cRab" score
  scores <- lapply(c(1:nrow(data)), function(i) {
    full_vec <- getPointVecHelper(i, mean_matrix)
    singlePointScore <- mean((1 - full_vec)^2)
    return(list(index = i, singlePointScore = singlePointScore))
  })
  final_df <- do.call(rbind.data.frame, scores)
  total_score <- mean(final_df$singlePointScore)

  final_df <- final_df |>
    mutate(singlePointContribution = singlePointScore / total_score)

  return(final_df)
}
```

Appendix E

GET POINT VECTOR HELPER

```
getPointVecHelper <- function(i, mean_matrix) {
  # Given a point i and the mean matrix, should
  # return vector of all values in the mean matrix corresponding to obs i
  # after removing all NAs and then taking absolute value

  row <- mean_matrix[i, ]
  col <- mean_matrix[, i]

  # We don't care about the order, just the values themselves
  res_vec <- c(row, col)
  res_vec <- abs(res_vec[!is.na(res_vec)])
  return(res_vec)

}
```

```
# resample_matrices <- resample_function(penguins,
#                                        number_of_resamples = 40,
#                                        proportion_resample = 0.8,
#                                        starting_seed = 100)
# m_matrix <- mean_matrix(resample_matrices)
#
#
```

```
# d <- point_contribution(penguins, m_matrix)
#
# test <- lapply(resample_matrices,
#        point_contribution,
#        data = penguins)
#
# mean_var_single <- do.call(rbind.data.frame, test) |>
#   group_by(index) |>
#   summarize(mean = mean(singlePointScore),
#             var = sd(singlePointScore)) |>
#   arrange(desc(var))
```